

Blockage-aware Multi-armed Bandits for Proactive Handover in mmWave Vehicular Networks

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I. INTRODUCTION

Autonomous Driving (AD), part of Intelligent Transportation Systems (ITS), aims to solve road problems like safety and fuel consumption. Connected Autonomous Vehicles (CAVs) use Vehicle-to-Everything (V2X) communications for coordinated functions. The European Telecommunications Standards Institute's Cooperative Awareness Message (CAM) [1] enables CAVs to share their status information.

CAV tasks demand high throughput and low latency. Fifth-generation (5G) millimeter-wave (mmWave) vehicular networks offer high data rates and reliability [2], but suffer from penetration loss and blockage sensitivity, resulting in small coverage and unstable connections [3]. The directional nature of mmWave requires Line-of-Sight (LoS) between transmitter and receiver for quality communication [4], which is challenging in high-mobility scenarios.

To stabilize mmWave connectivity under dynamic blockage, communication handover (HO) between Base Stations (BSs) is a common solution. However, conventional reactive HO schemes fail to meet real-time requirements for CAV applications due to passive decision-making during time-to-trigger (TTT). Vehicle speed causes frequent handovers, risking service disconnections. Therefore, systems enabling seamless, robust HO are essential for continuous connectivity.

In this work, we focus on the future blockage-aware proactive HO issue. In particular, our original contribution can be concluded as:

- This work includes an original attempt to design mmWave HO for V2X communication service under the high-mobility scenario that aims to avoid sudden outage of the communication signal due to vehicle blockages.
- The proposed proactive ML-assisted HO scheme is based on Contextual Multi-Armed Bandit (CMAB) that not only considers the vehicular kinematics, but also involves the predictive blockage impact concerning the presence of big vehicles.
- The comparison of the overall performance among the reactive HO schemes, the proactive schemes that are purely based on current observation, and the proposed method that considers the future blockage rate will be made.

II. RELATED WORK

Machine Learning (ML) is used in a predictive approach to improve the HO decision process, the base station discovery rate, and reduce the number of unnecessary HO [5]. Particularly, the CMAB theory, a sub-class of Reinforcement Learning (RL) problem that focuses on sequential decision-making under uncertainty, has demonstrated its effectiveness in HO management under the dynamic vehicular environments.

Reference [6] focuses on the proactive HO that predicts the next base station choice, with rewards focused on maximizing predicted SNR stability and minimizing unnecessary HO and service interruptions. However, the HO decision needs extra training effort, which takes advantage of the observed SNR histories. Authors in [7] developed a DDQN agent for 5G mmWave Dual-Active Protocol Stack (DAPS) handovers, learning when to trigger handovers to avoid sudden radio link failures under volatile mmWave channels. However, it is a reactive HO scheme based on prior signal degradation patterns, which does not explicitly predict future obstacles (e.g., motion tracking, prediction of moving blockages like vehicles).

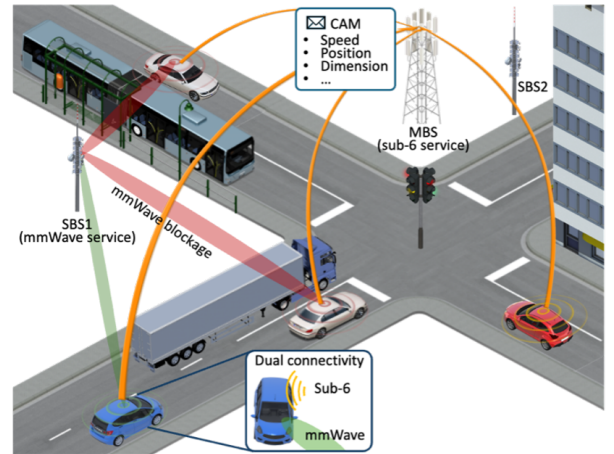


Fig. 1. Research Scenario

III. BLOCKAGE-AWARE MAB FOR PROACTIVE HO

A. System Overview

The research scenario is shown in Fig. 1. We examine a heterogeneous mmWave vehicular network with regions where a sub-6 GHz macro-Base Station (MBS) overlays multiple mmWave sub-BSs (SBSs). The MBS functions as

both an access point and a central processor for learning tasks, while SBSs enable advanced V2X applications. We assume varying traffic densities and large vehicles. Entering vehicles update their status via Cooperative Awareness Message (CAM) through V2I communication, providing input for mmWave blockage prediction and mitigation strategies.

This scenario is modeled as a CMAB problem with each SBS as an individual arm. The MBS, serving as a centralized learner, executes the following process (Fig. 2):

- **Context collection stage:** MBS collects vehicle CAMs in lower frequency bands, extracts context, and predicts future blockage rates based on trajectory forecasting. The context space is divided into uniform hypercubes containing classified context. Simultaneously, serving SBSs reports vehicle QoS.
- **HO detection stage:** Based on blockage status and current QoS, MBS triggers handover processes for cases including long future blockage duration or imminent significant QoS drops.
- **Execution and learning stage:** Before handover execution, MBS chooses to either explore (random arm selection) or exploit (select highest-reward arm). This trade-off enables system exploration while maintaining performance. As context inputs and reward tables update over time, MBS learns cross-scenario similarities for improved decision-making.

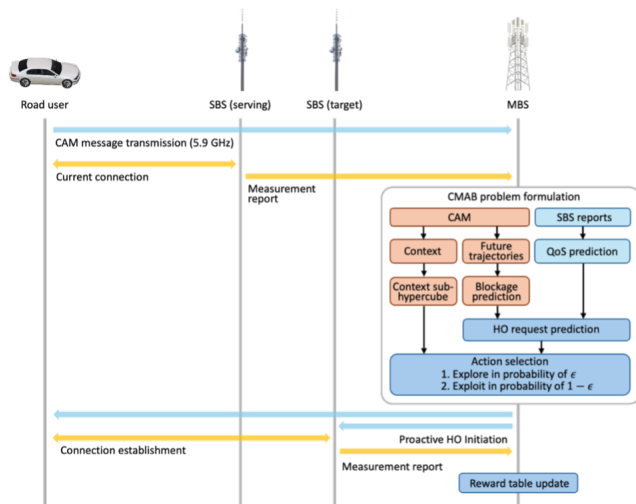


Fig. 2. Blockage-aware Proactive HO System Architecture

B. System Model

In the previously described MBS-governed mmWave vehicular network with multiple N_b SBSs, let $\mathcal{B} = \{1, \dots, j, \dots, N_b\}$ be the set of SBSs in a region. The MBS collects the vehicle dynamics through CAM via V2I communication at each time frame $t = 1, \dots, T$ in a finite time horizon $T \in \mathbb{N}$, where the vehicle set for each time frame is taken from $\mathbb{U}(t) = \{1, \dots, i, \dots, |\mathbb{U}(t)|\}$.

Therefore, each vehicle $v_{t,i}$ is characterized by a context vector $x_{t,i}$ taken from the X -dimensional context space $\mathcal{X} = [0, 1]^X$, where $X \in \mathbb{N}$. Each context dimension reflects certain

attributes that describe the vehicles, which can be discrete or continuous quantities, such as the velocity, acceleration, heading, and the appearance rate of big vehicles in the adjacent lane. Let $\gamma_{i,j}(t)$ denote the actual Signal-to-interference-plus-noise ratio (SINR) that vehicle $v_{t,i}$ (with context $x_{t,i}$) connecting to SBSs $j \in \mathcal{B}$ in time slot t .

In order to formulate the problem, consider the variable $y_{i,j}(t)$ as a binary decision variable that represents the SBSs selection. If the j th SBSs is selected in time slot t , then $y_{i,j}(t) = 1$, otherwise $y_{t,j} = 0$. Noted that each vehicle can connect to only one SBSs at time slot t .

Then, assume that the instantaneous transmission rate of vehicle i communicating to SBS j is:

$$R_{i,j}(t) = W \log_2 (1 + \gamma_{i,j}(t)) \quad (1)$$

In each period t , one vehicle can only communicate with one BS. Let $\beta_i(t)$ indicate the index of the serving BS for vehicle i . Therefore, the total system capacity is calculated as:

$$r(t) = \sum_{j \in \mathcal{B}} \sum_{i \in \mathbb{U}(t)} y_{i,j}(t) R_{i,j}(t) \quad (2)$$

Finally, the problem is formulated as maximizing Eq. 2.

IV. CONCLUSION

This work has proposed a CMAB-based HO scheme for the mitigation of dynamic blockages under an mmWave V2X communication scenario. The future work will focus on exploiting future information of vehicles for HO in a proactive manner. The use of the real-world dataset in Japan is expected.

V. ACKNOWLEDGEMENT

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